b) $\forall x(x\leq 2)$ $\quad \neg\forall x(x\leq 2) \equiv \exists x\,\neg(x\leq 2)$ (By De Morgan's laws of quantifiers)

$\equiv \exists x(x>2)$

c) $\exists x(x\geq 4)$ $\quad \neg\exists x(x\geq 4) \equiv \forall x\,\neg(x\geq 4)$ (By De Morgan's laws of quantifiers)

$\equiv \forall x(x<4)$

d) $\exists x(x<0)$ $\quad \neg\exists x(x<0) \equiv \forall x\,\neg(x<0)$ (By De Morgan's laws of quantifiers)

$\equiv \forall x(x\geq 0)$

e) $\forall x((x<-1)\vee(x>2))$ $\quad \neg\forall x((x<-1)\vee(x>2)) \equiv \exists x\,\neg((x<-1)\vee(x>2))$ (De Morgan's laws of quantifiers) $\equiv \exists x(\neg(x<-1)\wedge\neg(x>2))$ (2nd De Morgan's law) $\equiv \exists x((x\geq -1)\wedge(x\leq 2))$

f) $\exists x((x<4)\vee(x>7))$ $\quad \neg\exists x((x<4)\vee(x>7)) \equiv \forall x\,\neg((x<4)\vee(x>7))$ (De Morgan's laws of quantifiers) $\equiv \forall x(\neg(x<4)\wedge\neg(x>7))$

$\equiv \forall x((x\geq 4)\wedge(x\leq 7))$

37) Find a counterexample, if possible, to these universally quantified statements, where the domain for all variables consists of all integers.

a) $\forall x(x^2\geq x)$. Let $\neg\forall x(x^2\geq x)$ be True. By De Morgan's law of quantifiers $\exists x\,\neg(x^2\geq x)$ is True or, $\exists x(x^2<x)$ is True.

Let $p\in\mathbb{Z}$, $p^2<p$. $\Rightarrow p(p-1)<0$. Two cases arise: 1) $p>0$ and $(p-1)<0$ or, $p>0$ and $p<1$. No such integer exist 2) $p<0$ and $(p-1)>0$ or $p<0$ and $p>1$. This is a contradiction.

$\therefore$ p does not exist. $\therefore \exists x(x^2<x)$ is False. $\therefore \neg\forall x(x^2\geq x)$ is False and hence by defn. of negation and double-negation law, $\forall x(x^2\geq x)$ is True.

b) $\forall x(x>0 \vee x<0)$. $0\in\mathbb{Z}$. $\exists\, x=0$. $0>0$ is False $0<0$ is False

By defn. of OR, $(x>0 \vee x<0)$ is False for $x=0$.

$\therefore x=0$ is the counterexample

c) $\forall x(x=1)$. Any integer, apart from 1, makes this statement False.